

Ishan Acharya-Gangopadhyay

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EDUCATION

Michigan State University
B.S. Electrical Engineering

East Lansing, MI

Expected May 2029

EXPERIENCE

Lead Backend/Infrastructure Engineer | Wensura

January 2026 – Present

- Built B2B SaaS platform for battery materials research with 5 active enterprise customers; led all infrastructure, security, and backend engineering
- Built AWS infrastructure from scratch with Terraform, including compute, databases, storage, networking, and CDN
- Refactored a 3,000-line monolith into a clean, type-safe FastAPI package with an ORM layer and field-level encryption for sensitive data
- Engineered a multi-stage LLM peer review pipeline with iterative retrieval-augmented generation and real-time response streaming
- Implemented database-level multi-tenant data isolation; integrated multi-factor auth and Stripe billing
- Hardened CI/CD with automated secret and vulnerability scanning and pinned dependencies; grew test coverage from 27% → 82%
- Conducted security testing on the AI Copilot feature; found and fixed a critical remote-code-execution vulnerability, a path traversal bug, and gaps in output validation and prompt-injection defenses
- Diagnosed and fixed cross-tenant data isolation bugs in the RAG pipeline that could have let customers see each other's data

Simulation Lead | Spartan Autonomous Robotics (SAR)

2026 – Present

- Lead the Simulation and Sim-to-Real workstream on the Perception and Mapping sub-team; built a multi-stage pipeline that validates robot behavior in simulation before real-world testing, gated by automated checks
- Ported quadruped gait firmware from Python to Rust for consistency between simulation and hardware; built a web dashboard for running and reviewing simulation trials
- Built a scan-import pipeline bringing real-world terrain scans into simulation for realistic terrain and traversability testing
- Ported and implemented a Design of Experiments (DoE) framework for structured robotics simulation experiments and analysis

Expert Contributor (EC) | Snorkel AI and other AI research platforms

2026 – Present

- Authored and reviewed benchmark tasks used to evaluate AI coding agents, including test suites, reference solutions, and fairness/anti-cheat checks
- Replicated ML research papers into self-contained benchmark environments with reference solutions and detailed grading rubrics (25+ criteria each)
- Fixed bugs across Python libraries and task pipelines, and caught fabricated or hallucinated ("AI slop") output in submissions under review

Electrical Subteam Engineer | MSU Solar Racing

Fall 2025 – 2026

- Worked on the embedded dashboard UI for solar car Lysander; set up automated headless rendering to validate UI changes without physical hardware

Director of Software | FRC Team 4829 Titanium Tigers

2021 – 2025

- Built a full robot simulation (digital twin) in Java, including physics modeling, sensor simulation, and motor control algorithms
- Introduced a multithreaded pose estimation pipeline across variable camera count with automatic switching and teleport detection; delivered the team's first reliable vision system
- Trained and mentored 10–15 members per year in robot programming, mechanics, and control theory

PROJECTS

Homelab Infrastructure | Proxmox · Docker · Tailscale

2026 – Present

- Built a two-node home server setup running containerized workloads: self-hosted CI runners, GPU-enabled dev environments, and a personal multi-agent AI system
- Configured secure cross-network access and an isolated sandbox so automated builds can't affect the host system

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, Rust, C, JavaScript

Frameworks: FastAPI, React, Vite, LightRAG, pgvector, LVGL

Cloud & Infra: AWS (ECS Fargate, Lambda, RDS, S3, CloudFront, SQS, DynamoDB, ECR, EFS, VPC, IAM, Route53, Security Groups), Terraform, Docker, GitHub Actions

Databases: PostgreSQL, Redis, DynamoDB

Observability: OpenTelemetry, Grafana Cloud, Gitleaks, Checkov, Trivy

AI / ML: RAG pipelines, LLM orchestration, Ollama local inference, vector embeddings, SSE streaming

Hardware: ESP32-S3, PlatformIO, I2C/SPI, Arduino, Raspberry Pi, soldering & rework